

Light Up Your Imagination

Design Your Own Circuit Card





The Glowing Greeting Card

No messy wires! There is a secret lightning path hidden inside the paper — and it makes the light glow!

Meet Our Magic Friends



The Silver Coin — Battery

Flat top is plus (+), bumpy bottom is minus (-). This tiny coin pushes electricity all around!



The Twin-Leg Bulb — LED

It has two legs: long leg is plus (+) and short leg is minus (-). It lights up when connected right!



The Shiny Ribbon — Copper Tape

This sticky, shiny tape is the secret path for electricity. It carries power from the battery to the LED!

Path of electricity

- Electricity is like a tiny traveller — it needs a complete, unbroken loop to keep moving and light up your LED!
- Think of it like the Indian Railways. A train can only travel if the iron tracks are fully connected from start to finish. If there is even one broken piece of track, the train stops! ⚡🚂
- In our circuit card, the copper tape is the track. The battery is the engine. The LED is the station that lights up when the train arrives. Keep the loop connected — and watch the magic happen!





The Power Station

Meet the tiny engine that pushes electricity! The Coin Cell Battery has a flat top (+) and a bumpy bottom (-). It fits in your dad's wristwatch too!



The Light Maker — LED

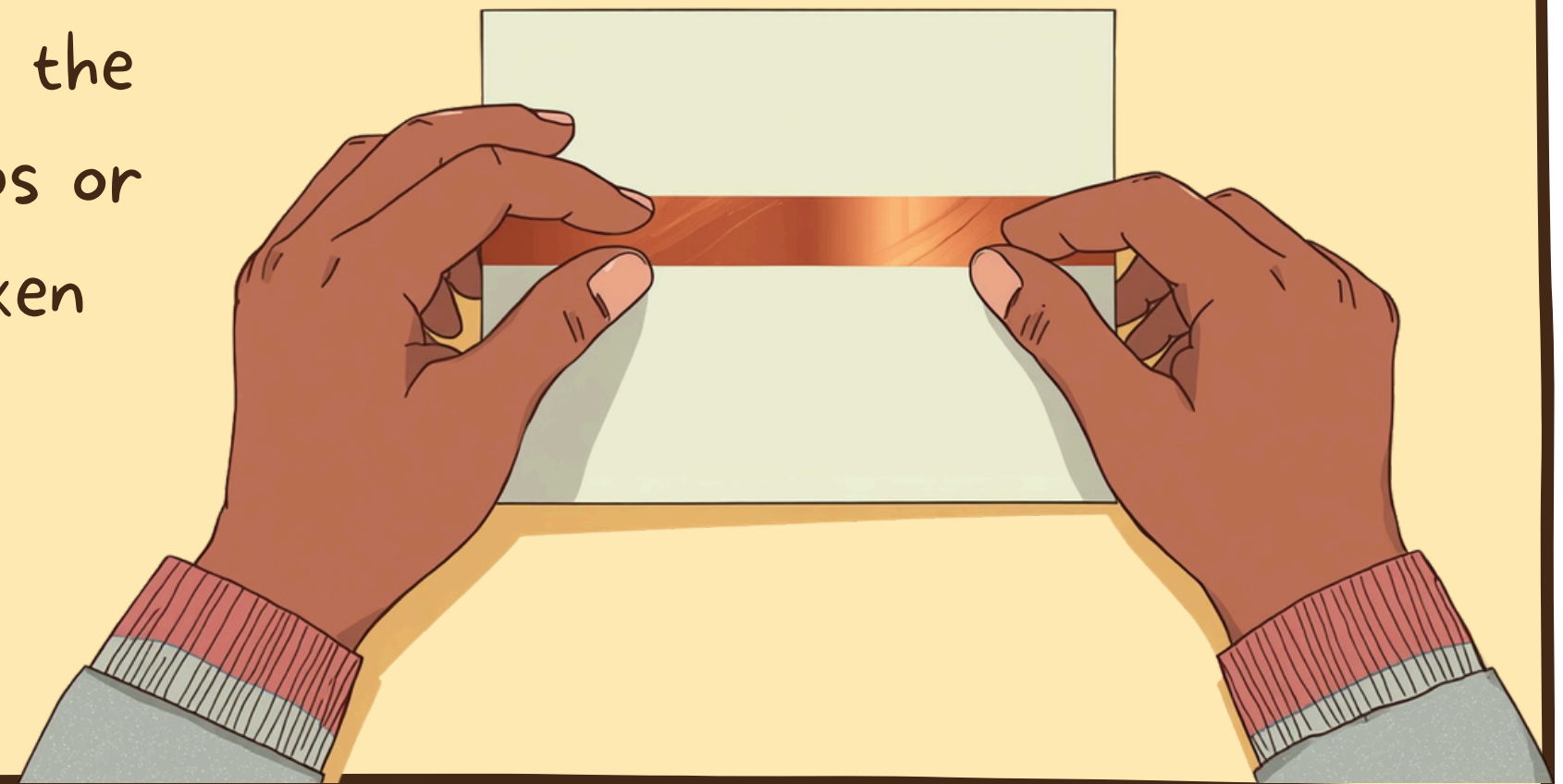
A tiny bulb with two legs:

- 👉 Long leg = Plus (+)
- 👉 Short leg = Minus (-)

Just like Diwali fairy lights that twinkle and glow!

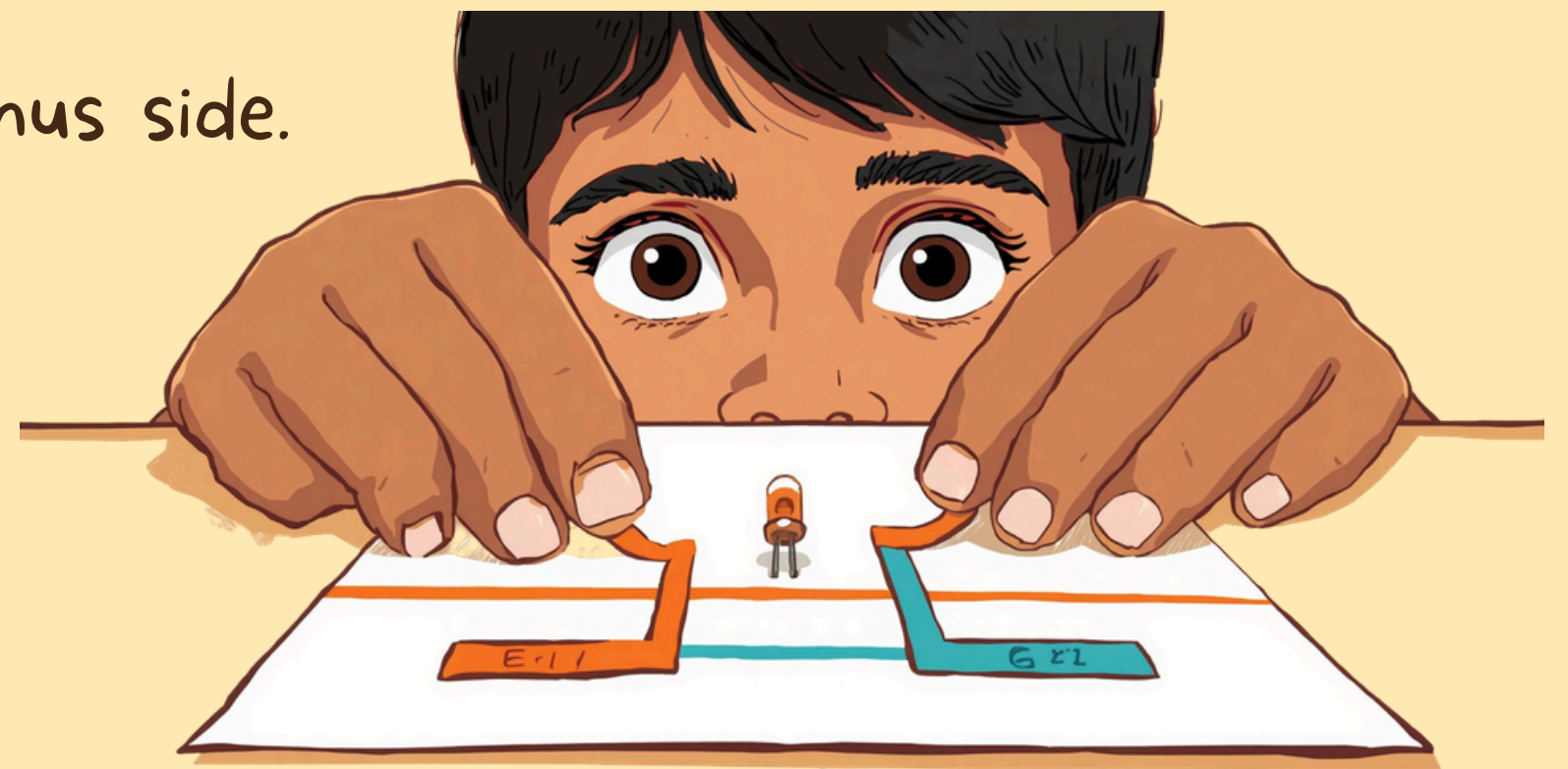
Step 1: Lay the Shiny Tape Path

- Take your copper tape and lay the red (+) path from the battery spot all the way to one leg of the LED. Then lay the blue (-) path from the other LED leg back to the battery.
- Leave a tiny 2mm gap at the top of the battery corner — this is your secret switch! Press the tape down firmly so it sticks flat. No bumps or tears — electricity needs a smooth, unbroken road to travel on!



Step 2: Place the Twin-Leg Bulb

- Now it's time to place your LED — the tiny light maker!
Carefully bend both legs flat against the card.
- The long leg (+) goes on the red tape path. The short leg (—) goes on the blue tape path. Press each leg down gently and tape over them to hold in place.
- Remember: long leg = plus side, short leg = minus side.
Getting this right makes your bulb glow!



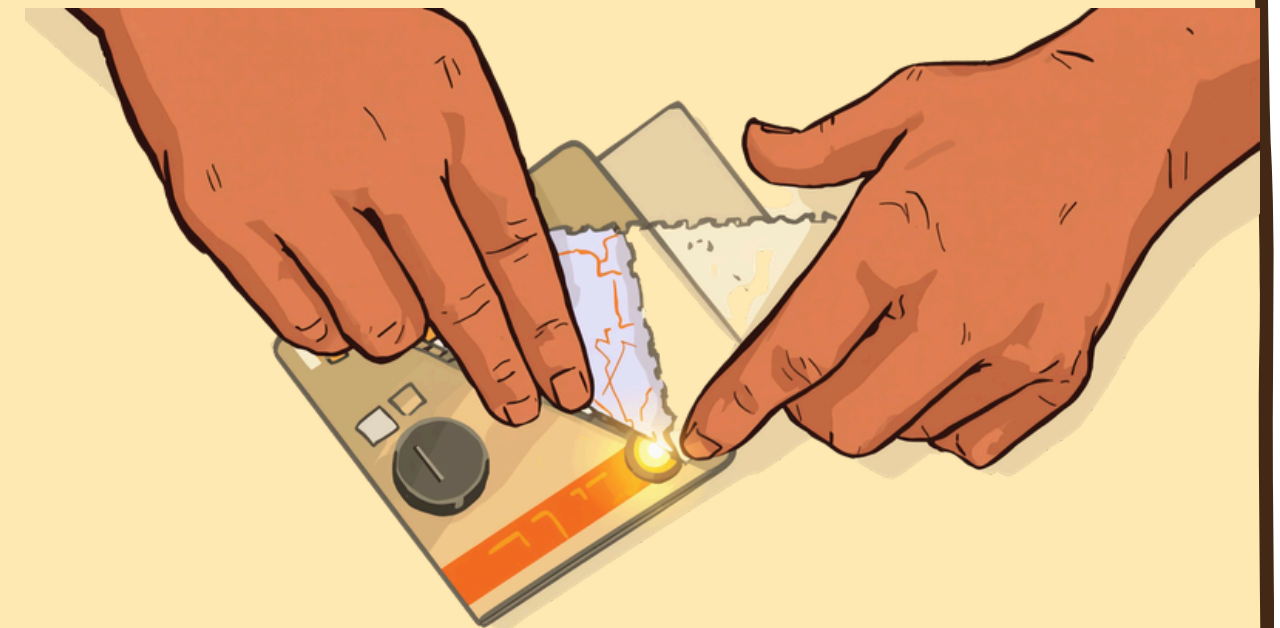
Step 3: Slide in the Silver Coin

- Time to add the power! The Silver Coin is our CR2032 coin cell battery — the tiny engine that pushes electricity through our card.
- Slide the battery into the bottom corner of your card. Make sure the flat side (the + side) faces UP. The bumpy side faces DOWN. Press the folded corner gently over the battery to hold it in place. You will see the LED glow as the circuit connects!



Step 4: Make the Switch Corner

- Fold the bottom corner of your card over the battery. This presses the copper tape onto the flat (+) side of the coin cell battery!
- When the corner is folded down, the circuit is complete — electricity flows from the battery, along the tape path, through the LED, and back again. Your LED will light up! Hold the corner down firmly to keep the light glowing.
- This folded flap is your switch. Lift it to turn the light OFF. Press it down to turn it ON. You just built a real working switch! ✨





Creative Theme Framing

Draw around your glowing LED to make it magical! Try a Diya flame, a Vande Bharat train headlight, or a shining star. What will YOU imagine?

Doctor-Check Game



✓ Check A: LED legs reversed?

Is the long leg on the red (+) tape path? Is the short leg on the blue (—) path? Swap and try again!

✓ Check B: Is the tape unbroken?

Look closely at your copper tape path. Any tears or gaps? Press the tape down firmly or add a new piece!

✓ Fix it and watch the light glow!

Once you fix the problem, press the switch corner and see your LED light up bright. You are a Circuit Doctor!



Celebrate Our Young Engineers!

You are now Hardware Engineers! Hold up your glowing card with pride — you built a real circuit! ✨



Thank You & Happy Making!

Share your glowing card with family and friends!